

专题4. 高阶导数中的解题方法.

- 考法一. 找规律法.
- 考法二. 莱布尼兹求导公式.
- 考法三. 泰勒展开法. (重要) ★★

考法四. 数学归纳法 (选学)

考法五. 最难考法: 利用莱布尼兹求导公式. 构造递推关系 (选学)

不讲!
只发手稿



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考法-找规律法.

$$y^n \quad y^{(n)}$$

$$y = f(x).$$

$$y = f(x+1)$$

以下公式无需记忆. 都可以用找规律法轻松推出 (其中 a, b 为常数. 且 $a \neq 0$).

$$(e^{ax+b})^{(n)} = a^n \cdot e^{ax+b}$$

$$(e^x)^{(n)} = e^x$$

$$(\sin x)' = \cos x = \sin(x + \frac{\pi}{2}) \quad (\cos x)' = -\sin x$$

$$= \cos(x + \frac{\pi}{2})$$

$$[\sin(ax+b)]^{(n)} = a^n \sin(ax+b + \frac{\pi}{2} \cdot n)$$

$$(\sin x)^{(n)} = \sin(x + \frac{\pi}{2} \cdot n)$$

$$[\cos(ax+b)]^{(n)} = a^n \cos(ax+b + \frac{\pi}{2} \cdot n)$$

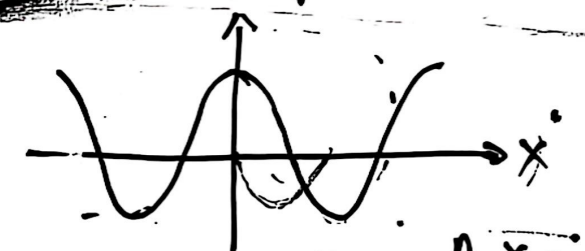
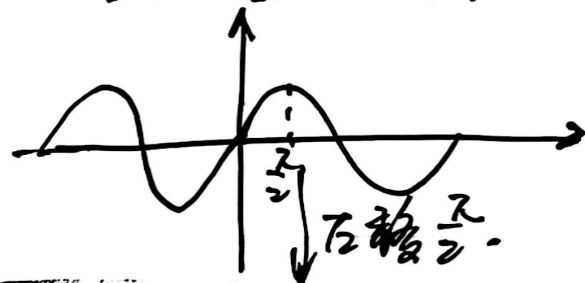
$$(\cos x)^{(n)} = \cos(x + \frac{\pi}{2} \cdot n)$$

$$[\ln(ax+b)]^{(n)} = (-1)^{n-1} \cdot a^n \cdot (n-1)! \cdot \frac{1}{(ax+b)^n}$$

$$(\ln x)^{(n)} = \frac{(-1)^{n-1} (n-1)!}{x^n}$$

$$(\frac{1}{ax+b})^{(n)} = (-1)^n \cdot a^n \cdot n! \cdot \frac{1}{(ax+b)^{n+1}}$$

$$(\frac{1}{x})^{(n)} = \frac{(-1)^n \cdot n!}{x^{n+1}}$$



$$y^{(n)} = (\sqrt{2})^n e^x \sin(x + \frac{\pi}{4} \cdot n)$$

例1. (1). $y = e^x \sin x$. 求 $y^{(n)}$; (2). $y = \frac{1}{2x+3}$. 求 $y^{(n)}$.

解: (1). $y = e^x \sin x + e^x \cos x = e^x \cdot \sqrt{2} \sin(x + \frac{\pi}{4}) \Rightarrow y'' = \sqrt{2} \cdot [e^x \sin(x + \frac{\pi}{4}) + e^x \cos(x + \frac{\pi}{4})] = (\sqrt{2})^2 e^x \sin(x + \frac{\pi}{4} \cdot 2)$

(2). $y = \frac{1}{2x+3} = (2x+3)^{-1} \Rightarrow y' = (-1) \cdot (2x+3)^{-2} \cdot 2 \Rightarrow y'' = (-1) \cdot (-2) \cdot (2x+3)^{-3} \cdot 2^2 \Rightarrow y''' = (-1) \cdot (-2) \cdot (-3) \cdot (2x+3)^{-4} \cdot 2^3$

$\Rightarrow y^{(n)} = (-1)(-2) \cdots (-n) (2x+3)^{-(n+1)} \cdot 2^n \cdot \frac{1}{2} \cdot 2^n \Rightarrow y^{(n)} = \frac{(-1)^n \cdot n!}{2^{n+1}} \cdot 2^n$

例2. 设 $f(x)$ 为 $\frac{1}{x}$ 的幂函数. $f'(x) = f^2(x)$, $f(0) = 2$. 求 $f^{(n)}(0)$

[解]: $f' = f^2 \Rightarrow f'' = 2f \cdot f' = 2f^3 \Rightarrow f''' = 2 \cdot 3 f^2 \cdot f' = 2 \cdot 3 f^4 \Rightarrow f^{(4)} = 1 \cdot 2 \cdot 3 \cdot 4 f^3 \cdot f' = 1 \cdot 2 \cdot 3 \cdot 4 \cdot f^5$

$\Rightarrow f^{(n)} = n! f^{n+1} \quad \text{令 } x=0 \Rightarrow f^{(n)}(0) = n! f^{n+1}(0) = 2^{n+1} \cdot n! \quad \checkmark$

考法二: 莱布尼兹求导公式 $(a+b)^n = C_n^0 a^n b^0 + C_n^1 a^{n-1} b^1 + \dots + C_n^n a^0 b^n \quad \checkmark$

因为多项式, 每导一次, 就少一次!

$(u \cdot v)^{(n)} = C_n^0 u \cdot v^{(n)} + C_n^1 u' \cdot v^{(n-1)} + C_n^2 u'' \cdot v^{(n-2)} + \dots + C_n^n u^{(n)} \cdot v$. 特别适用于 u 和 v 中至少有一个为多项式

(提示: 以下4道题均有更好的解法. 这里只是为刻意展示这个方法而已).

例3. (2000年). $y = x^2 \ln(x+1)$. 求 $y^{(n)}(0)$. ($n \geq 3$)

[解]: $y^{(n)} = C_n^0 x^2 [\ln(x+1)]^{(n)} + C_n^1 2x [\ln(x+1)]^{(n-1)} + C_n^2 2 \cdot [\ln(x+1)]^{(n-2)}$

对于 $[\ln(x+1)]^{(n-2)} = \frac{(-1)^{n-1} (n-3)!}{(x+1)^{n-2}} \Rightarrow y^{(n)}(0) = \frac{n(n-1)}{2} \cdot (-1)^{n-1} \cdot \frac{(n-3)!}{1} = \frac{(-1)^{n-1} \cdot n!}{(n-2)!} \quad \checkmark$

例4. 设 $f(x) = (x-1)^n \cdot \boxed{x^{2n} \sin \frac{\pi}{2} x}$. 求 $f^{(n)}(1)$.

解: 要有整体的思想. 把 $(x-1)^n$ 以外的函数. 看成一个整体. 令 $\underline{g(x) = x^{2n} \sin \frac{\pi}{2} x}$.

$$f(x) = (x-1)^n \cdot g(x) \Rightarrow f^{(n)} = \underline{C_n^0 \cdot \boxed{(x-1)^n} g^{(n)}(x)} + C_n^1 \cdot n(x-1)^{n-1} \cdot g^{(n-1)}(x) + \dots + \underline{C_n^{n-1} \cdot n! (x-1) \cdot g'(x)} + \boxed{C_n^n n! g(x)}$$

$$f^{(n)}(1) = 0 + 0 + \dots + 0 + C_n^n \cdot n! g(1) \Rightarrow f^{(n)}(1) = n! g(1) = n! \cdot 1^{2n} \cdot \sin \frac{\pi}{2} = \boxed{n!} \checkmark$$

变题. 设 $f(x) = (x^2 - 3x + 2)^n \cdot \cos(\frac{\pi}{16} x^2)$. 求 $f^{(n)}(2)$.

解: 方法同上. $f = (x-2)^n \cdot \boxed{(x-1)^n \cos \frac{\pi}{16} x^2} = (x-2)^n \cdot g(x)$

$$f^{(n)} = C_n^0 \cdot \underline{(x-2)^n} g^{(n)}(x) + C_n^1 \cdot n(x-2)^{n-1} \cdot g^{(n-1)}(x) + \dots + C_n^n \cdot n! g(x).$$

令 $x=2 \Rightarrow f^{(n)}(2) = n! g(2) = n! \cdot 1^n \cdot \cos(\frac{\pi}{16} \cdot 4) = n! \cdot \frac{\sqrt{2}}{2} \checkmark$



例5. 设 $f(x) = (x-a)^n \cdot y(x)$. 其中 $y(x)$ 在 $x=a$ 的某邻域内有 $n-1$ 阶连续导数. 求 $f^{(n)}(a)$.

[解]: 错解 — $f^{(n)}(x) = C_n^0 \cdot (x-a)^n \cdot \cancel{y^{(n)}(x)} + C_n^1 \cdot n(x-a)^{n-1} \cdot y^{(n-1)}(x) + \dots + \boxed{C_n^n \cdot n! \cdot y(x)}$
 令 $x=a \Rightarrow f^{(n)}(a) = C_n^n \cdot n! \cdot y(a) = \underline{n! y(a)}$. 结果正确. 过程错!

正确解法: $f^{(n-1)}(x) = C_{n-1}^0 \cdot (x-a)^n \cdot y^{(n-1)}(x) + C_{n-1}^1 \cdot n(x-a)^{n-1} \cdot y^{(n-2)}(x) + \dots + C_{n-1}^{n-1} \cdot n! \cdot (x-a) \cdot y(x)$

$$f^{(n)}(a) = \lim_{x \rightarrow a} \frac{f^{(n-1)}(x) - f^{(n-1)}(a)}{\underbrace{(x-a)}} = \underbrace{0 + 0 + \dots + 0}_{n-1 \uparrow 0} + C_{n-1}^{n-1} \cdot n! \cdot y(a) = \underline{\underline{n! y(a)}}$$

考法三. 泰勒展开法.

该方法适用于求具体一点处的 n 阶导 (但无法求 n 阶导函数). 其方法为——

step1. 先将题中的 $f(x)$ 泰勒展开为 $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n + \dots$ 的样子.

step2. 又由 $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$

对比相同次方项的系数. 得 $\frac{f^{(n)}(0)}{n!} = a_n$. 解得 $f^{(n)}(0) = n! \cdot a_n$ ✓.

(总结: 具体展开, 抽象展开, 对比系数. 得出答案!).

回顾刚才的2000年真题—— $y = \sqrt{x} \ln(1+x)$. 求 $y^{(n)}(0)$.

[解]: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n+1} \frac{x^n}{n} + \dots$

$$\Rightarrow y = x^2 \left[x - \frac{x^2}{2} + \dots + (-1)^{n+1} \frac{x^{n-2}}{n-2} + \dots \right] \Rightarrow \frac{(-1)^{n+1}}{n-2} = \frac{y^{(n)}(0)}{n!} \Rightarrow y^{(n)}(0) = \frac{(-1)^{n+1} \cdot n!}{n-2}$$

$$\frac{(-1)^{n-3}}{n-2} = (-1)^{n+1}$$



例 6. 设 $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ 求 $f^{(5)}(0)$ 和 $f^{(6)}(0)$

[解]: $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

故 $\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots \quad (x \neq 0)$

故 $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases} = \begin{cases} 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots, & x \neq 0 \\ 1, & x = 0 \end{cases} \Rightarrow$

$0 \cdot x^5$

$\uparrow$
 $0 = \frac{f^{(5)}(0)}{5!} \Rightarrow f^{(5)}(0) = 0$

$-\frac{1}{7!} = \frac{f^{(6)}(0)}{6!} \Rightarrow f^{(6)}(0) = -\frac{1}{7} \checkmark$

$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - \dots$

例 7. (2007年). 设 $y = \frac{1}{2x+3}$, 求 $y^{(n)}(0) = \underline{\hspace{2cm}}$

[解]: $y = \frac{1}{3} \cdot \frac{1}{1 + (\frac{2}{3}x)} = \frac{1}{3} \left[1 - \frac{2}{3}x + \left(\frac{2}{3}x\right)^2 - \left(\frac{2}{3}x\right)^3 + \dots + (-1)^n \left(\frac{2}{3}x\right)^n + \dots \right]$

$\Rightarrow \frac{1}{3} \cdot (-1)^n \cdot \left(\frac{2}{3}\right)^n = \frac{y^{(n)}(0)}{n!} \Rightarrow y^{(n)}(0) = \frac{(-1)^n 2^n \cdot n!}{3^{n+1}} \checkmark$

例题2. 设 $y = e^{\sqrt{2}x} \sin x$. 求 $y^{(5)}(0)$.

解: $y = \left[1 + \sqrt{2}x + \frac{(\sqrt{2}x)^2}{2!} + \frac{(\sqrt{2}x)^3}{3!} + \frac{(\sqrt{2}x)^4}{4!} + \dots \right] \cdot \left[x - \frac{x^3}{6} + \frac{x^5}{5!} - \dots \right]$

x^5 前的系数: $\frac{1}{5!} + \left(\frac{2}{2}\right) \cdot \left(-\frac{1}{6}\right) + \frac{4}{4!} \cdot 1 = \frac{y^{(5)}(0)}{5!} \Rightarrow y^{(5)}(0) = 1 - \frac{5!}{6} + \frac{5!}{3!} = 1$ ✓

$Q = x + x^2$

$\frac{1}{1+Q} = 1 - Q + Q^2 - Q^3 + Q^4 - \dots$

$1 - x^3 = (1-x)(1+x+x^2)$

$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

例题1. 设 $f(x) = \frac{1}{1+x+x^2}$. 求 $f^{(10)}(0)$

解: 第一步最难! $f = \frac{1-x}{(1-x)(1+x+x^2)} = \frac{1-x}{1-x^3} = \frac{1}{1-x^3} + (-x) \cdot \frac{1}{1-x^3}$

$= [1 + x^3 + x^6 + x^9 + x^{12} + \dots] + (-x) \cdot [1 + x^3 + x^6 + x^9 + x^{12} + \dots]$

x^{10} 前系数: $(-1) = \frac{f^{(10)}(0)}{10!} \Rightarrow f^{(10)}(0) = -10!$ ✓



例8. 设 $f(x) = \ln(\sin x + \sqrt{1 + \sin^2 x})$. 求 $f^{(3)}(0)$, $f^{(4)}(0)$

$$y = \ln(x + \sqrt{1+x^2}) = x - \frac{1}{6}x^3 + \dots$$

解: 首先, 由于 $f(x)$ 是奇函数, 故 $f^{(2n)}(x)$ 全都是奇函数, 故 $f^{(2n)}(0) = 0$. 故 $f^{(4)}(0) = 0$.

故只需求 $f^{(3)}(0)$

$$f(x) = \sin x - \frac{1}{6}\sin^3 x + \dots = \left(x - \frac{x^3}{6}\right) - \frac{1}{6}x^3 + \dots \Rightarrow -\frac{1}{3} = \frac{f^{(3)}(0)}{3!} \Rightarrow f^{(3)}(0) = -2$$

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}} = e \cdot e^{\lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1}} = e \cdot e^1 = e^2$$

变题: 设 $f(x) = e^{\frac{1}{1+x}}$ 求 $f^{(4)}(0)$ 和 $f^{(5)}(0)$ 改为求 $f^{(3)}(0)$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

$$\begin{aligned} \text{解: } f &= e \cdot e^{\frac{1}{1+x}-1} = e \cdot e^{\frac{-x}{1+x}} \sim (-x) \\ &= e \cdot \left[1 + \frac{-x}{1+x} + \frac{1}{2!} \left(\frac{-x}{1+x} \right)^2 + \frac{1}{3!} \left(\frac{-x}{1+x} \right)^3 + \dots \right] \\ &= e \left[1 + \underline{(-x)} + \frac{x^2}{2} + \dots \right] \end{aligned}$$

$$x^3 \text{ 前的系数: } e \cdot \left[\underline{(-1)} + \frac{1}{2} \underline{(-2)} - \frac{1}{6} \right] = \frac{f^{(3)}(0)}{3!} \Rightarrow f^{(3)}(0) = 6e \cdot \left[-2 - \frac{1}{6} \right] = \underline{\underline{-13e}}$$

例. 设 $f(x) = (x-1)^n \cdot x^{2n} \cdot \sin \frac{\pi}{2}x$. 求 $f^{(n)}(1)$.

[回顾]: 上-次课. $f(x) = (1-\cos x) \cdot (2-\cos 2x) \cdot (3-\cos 3x) \cdots (n-\cos nx)$. 求 $f^{(n)}(0)$ ✓

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \frac{1}{2} \cdot 1 \cdot 2 \cdots (n-1) = \frac{(n-1)!}{2}$$

$$\text{又} \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \lim_{x \rightarrow 0} \left(f(x) + f'(x)x + \frac{f''(0)}{2!}x^2 + o(x^2) \right) \Rightarrow \begin{cases} f(0)=0 \\ f'(0)=0 \\ \frac{f''(0)}{2!} = \frac{(n-1)!}{2} \end{cases} \Rightarrow f''(0) = (n-1)!$$

[本题]: $\lim_{x \rightarrow 1} \frac{f(x)}{(x-1)^n} = \lim_{x \rightarrow 1} x^{2n} \cdot \sin \frac{\pi}{2}x = 1$

$$\text{又} \lim_{x \rightarrow 1} \frac{f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \cdots + \frac{f^{(n)}(1)}{n!}(x-1)^n + o[(x-1)^n]}{(x-1)^n} = 1 \Rightarrow \begin{cases} f(1) = f'(1) = \cdots = f^{(n-1)}(1) = 0 \\ \frac{f^{(n)}(1)}{n!} = 1 \end{cases} \Rightarrow f^{(n)}(1) = n!$$

例题1. 设 $f(x) = (x^2-3x+2)^n \cos \frac{\pi}{6}x$. 求 $f^{(n)}(2)$

$$\text{[解]: } \lim_{x \rightarrow 2} \frac{f(x)}{(x-2)^n} = \lim_{x \rightarrow 2} (x-1)^n \cos \frac{\pi x}{6} = \frac{\sqrt{2}}{2} = \lim_{x \rightarrow 2} \frac{f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + o[(x-2)^2]}{(x-2)^n} \Rightarrow \begin{cases} f(2) = \cdots = f^{(n-1)}(2) = 0 \\ \frac{f^{(n)}(2)}{n!} = \frac{\sqrt{2}}{2} \end{cases}$$

例题2. 设 $y = (x^2-1)^n$. 求 $y^{(n)}(-1)$ 和 $y^{(n)}(1)$

$$\text{[解]: } y = (x-1)^n (x+1)^n \Rightarrow \lim_{x \rightarrow 1} \frac{y}{(x-1)^n} = \lim_{x \rightarrow 1} (x+1)^n = 2^n \Rightarrow \begin{cases} y(1) = y'(1) = \cdots = y^{(n-1)}(1) = 0 \\ \frac{y^{(n)}(1)}{n!} = 2^n \Rightarrow y^{(n)}(1) = n! \cdot 2^n \end{cases}$$



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考法四. 数学归纳法 (选学)

(本方法主要用于证明题)

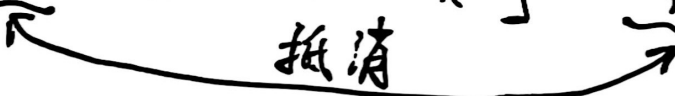
例10. 证明: $[x^{n+1} \cdot f(\frac{1}{x})]^{(n)} = \frac{(-1)^n}{x^{n+1}} \cdot f^{(n)}(\frac{1}{x})$

[解]: (1) 验证初值: $n=1$ 时. 左边 = $[f(\frac{1}{x})]' = -\frac{1}{x^2} \cdot f'(\frac{1}{x})$; 右边 = $\frac{(-1)}{x^{1+1}} f'(\frac{1}{x})$. 故成立.

(2). 假设该结论对 n 成立. 即 $[x^{n+1} \cdot f(\frac{1}{x})]^{(n)} = \frac{(-1)^n}{x^{n+1}} f^{(n)}(\frac{1}{x})$

(3). $[x^n \cdot f(\frac{1}{x})]^{(n+1)} = \left([x \cdot \underbrace{x^{n+1} f(\frac{1}{x})}^{(n)}]^{(n)} \right)' \xrightarrow{\text{莱布尼兹公式}} \left[C_n^0 \cdot x \cdot (x^{n+1} f(\frac{1}{x}))^{(n)} + C_n^1 \cdot 1 \cdot (x^{n+1} f(\frac{1}{x}))^{(n-1)} \right]'$

用(2)的假设 $\left[x \cdot \frac{(-1)^n}{x^{n+1}} f^{(n)}(\frac{1}{x}) \right]' + n \cdot \frac{(-1)^n}{x^{n+1}} f^{(n)}(\frac{1}{x}) = \cancel{\frac{(-1)^n}{x^{n+1}} f^{(n)}(\frac{1}{x})} \left[\right.$
 $= (-1)^n \cdot \left[\frac{1}{x^n} f^{(n)}(\frac{1}{x}) \right]' + n \cdot \frac{(-1)^n}{x^{n+1}} f^{(n)}(\frac{1}{x})$

$= (-1)^n \cdot \left[(-n) \cdot \frac{1}{x^{n+1}} f^{(n)}(\frac{1}{x}) + \frac{1}{x^n} \cdot f^{(n+1)}(\frac{1}{x}) \cdot (-\frac{1}{x^2}) \right] + n \cdot \frac{(-1)^n}{x^{n+1}} f^{(n)}(\frac{1}{x})$


$= \frac{(-1)^{n+1}}{x^{n+2}} \cdot f^{(n+1)}(\frac{1}{x})$. 说明该结论对 $n+1$ 也成立.

故由数学归纳法. 该结论对 $\forall n$ 均恒成立. 证毕!

考法五. 最难考法: 利用莱布尼兹求导公式. 构造线性递推 (选学).

例10. 已知 $y = \arctan x$. 求 $y^{(n)}(0)$

解: $y' = \frac{1}{1+x^2} \Rightarrow (1+x^2) \cdot y' = 1$. 两边同时求 n 阶导. 对右边使用莱布尼兹求导公式.

$$C_n^0 \cdot (1+x^2) \cdot y^{(n+1)} + C_n^1 \cdot 2x \cdot y^{(n)} + C_n^2 \cdot 2 \cdot y^{(n-1)} = 0.$$

令 $x=0 \Rightarrow y^{(n+1)}(0) + n(n-1) \cdot y^{(n-1)}(0) = 0 \Rightarrow y^{(n+1)}(0) = -n(n-1) y^{(n-1)}(0)$. 这是隔次递推. 要对 n 奇偶.

$$\begin{aligned} (1) \quad y^{(2n+1)}(0) &= -2n(2n-1) \cdot y^{(2n-1)}(0) = [-2n(2n-1)] \cdot [-(2n-2)(2n-3)] y^{(2n-3)}(0) \\ &= \dots = [-2n(2n-1)] \cdot [-(2n-2)(2n-3)] \cdots [-2 \cdot 1] \cdot (y'(0)) = 1 \\ &= (-1)^n \cdot (2n)! \end{aligned}$$

(2). 显然 $y^{(2n)}(0) = 0$ (奇偶性).

$$\text{综上. } y^{(n)}(0) = \begin{cases} (-1)^k \cdot (2k)!, & n=2k+1 \\ 0, & n=2k \end{cases} \quad (k=0, 1, 2, \dots).$$



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类题. 设 $y = \arcsin x$. 求 $y^{(n)}(0)$

[解]: $y' = \frac{1}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2} \cdot y' = 1$, 再求导. $\frac{-2x}{2\sqrt{1-x^2}} \cdot y' + \sqrt{1-x^2} \cdot y'' = 0 \Rightarrow -xy' + (1-x^2)y'' = 0$

(得到了这个线性微分方程以后), 再求 n 阶导. 用莱布尼兹求导公式. 得

$$-[C_n^0 \cdot x \cdot y^{(n+1)} + C_n^1 \cdot 1 \cdot y^{(n)}] + [C_n^0 \cdot (1-x^2) \cdot y^{(n+2)} + C_n^1 \cdot (-2x) \cdot y^{(n+1)} + C_n^2 \cdot (-2) \cdot y^{(n)}] = 0$$

令 $x=0 \Rightarrow -n \cdot y_{(0)}^{(n)} + y_{(0)}^{(n+2)} - n(n-1) \cdot y_{(0)}^{(n)} = 0 \Rightarrow y_{(0)}^{(n+2)} = n^2 \cdot y_{(0)}^{(n)}$. 奇偶性:

(1) $y_{(0)}^{(2n)} = 0$ (奇偶性)

(2). $y_{(0)}^{(2n+1)} = (2n-1)^2 \cdot y_{(0)}^{(2n-1)} = (2n-1)^2 \cdot (2n-3)^2 \cdot y_{(0)}^{(2n-3)} = \dots = (2n-1)^2 (2n-3)^2 \dots 3^2 \cdot 1^2 \cdot y_{(0)}^{(1)}$
 $= [(2n-1)!!]^2 \cdot 1$

综上: $y_{(0)}^{(n)} = \begin{cases} 0, & n=2k \\ [(2k-1)!!]^2, & n=2k+1 \end{cases}$